Lecture 3 Notes - Characteristics of Time Series

Liberty Hamilton

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- **Reading**: Chapter 1.3-1.7 – Shumway and Stoffer

1 Review / basic concepts

- Stochastic process
- White noise
- Moving average
- Autoregression
- Random walk (w/ and w/o drift)
- Signals in noise

2 This week - Measures of dependence

- Mean function
- Autocovariance
- Autocorrelation
- Cross-covariance
- Cross-correlation
- Stationarity
- Multidimensional time series
- Random number generation

3 Mean and variance

Last week and in your lab, we saw the concept of how mean and variance differ for a white noise process, a random walk, and a random walk with drift. The *mean* and *variance functions* of a time series are useful descriptors because they helps us determine something about the drift and the spread of the data that we should expect over time.

The *mean function* is defined as $\mu_{xt} = \mathbb{E}(x_t)$

The *variance function* is $\sigma_t^2 = \text{Var}(x_t) = \mathbb{E}[(x_t - \mu_t)^2]$.

Let's revisit some examples from before:

3.1 White noise

For a white noise time series, $\mu_{wt} = \mathbb{E}(w_t) = 0$ for all t. $\text{Var}(w_t) = 1$ (for a Gaussian white noise series).

3.2 Moving average

What if we apply a 3-point moving average? Although this induces some correlation structure, it actually does not change the mean function at all:

$\mu_{vt} = \mathbb{E}(v_t) = \frac{1}{3}[\mathbb{E}(w_{t-1}) + \mathbb{E}(w_t) + \mathbb{E}(w_{t+1})] = 0$

3.3 Random walk with drift

For the random walk with drift, the mean function is just the line $\mu_{xt} = \delta t + \sum_{j=1}^{t} E(w_j) = \delta t$.

3.4 Signal plus noise

And for a signal plus noise (as an example, we'll use this sinusoid):

$$\begin{aligned} \mu_{xt} = \mathbb{E}(x_t) &= \mathbb{E}[A\cos(2\pi\omega t + \phi) + w_t] \\ &= \mathbb{E}[A\cos(2\pi\omega t + \phi)] + \mathbb{E}[w_t] \\ &= \mathbb{E}[A\cos(2\pi\omega t + \phi)] \end{aligned} \tag{1}$$

3.5 Autocovariance

What if we instead want to know something about the dependence between two points s and t within the same time series. We'll call this *autocovariance*, γ.

$\gamma_x(s,t) = \text{cov}(x_s, x_t) = \mathbb{E}[(x_s - \mu_s)(x_t - \mu_t)]$

When $s = t$, we have:

$\gamma_x(t,t) = \mathbb{E}[(x_t - \mu_t)^2] = \text{Var}(x_t)$

For white noise, there should be no dependence between differing time points. By definition, $\mathbb{E}(w_t) = 0$ and:

$$\gamma_w(s,t) = \mathrm{cov}(w_s, w_t) = \begin{cases} \sigma_w^2, & \text{if } s = t, \\ 0, & s \neq t \end{cases} \tag{2}$$

3.6 Covariance of linear combinations

If we have random variables $U = \sum_{j=1}^{m} a_j X_j$ and $V = \sum_{k=1}^{r} b_k Y_k$ that are linear combinations of (finite variance) random variables X_j and Y_k, then the covariance of these is:

$\mathrm{cov}(U,V) = \sum_{j=1}^{m}\sum_{k=1}^{r} a_j b_k \,\mathrm{cov}(X_j, Y_k)$

Also, $\mathrm{var}(U) = \mathrm{cov}(U,U)$.

So how can we use this? Now we can try this for the moving average example.

$$\begin{aligned} \gamma_v(s,t) = \mathrm{cov}(v_s, v_t) &= \mathrm{cov}(\tfrac{1}{3}(w_{s-1} + w_s + w_{s+1}), \tfrac{1}{3}(w_{t-1} + w_t + w_{t+1})) \\ &= \tfrac{1}{9}\,\mathrm{cov}(w_{s-1} + w_s + w_{s+1}, w_{t-1} + w_t + w_{t+1}) \end{aligned} \tag{3}$$

When $s = t$, we have:

$$\begin{aligned} \gamma_v(t,t) &= \mathrm{cov}(v_t, v_t) \\ &= \mathrm{cov}(\tfrac{1}{3}(w_{t-1} + w_t + w_{t+1}), \tfrac{1}{3}(w_{t-1} + w_t + w_{t+1})) \\ &= \tfrac{1}{9}(\mathrm{cov}(w_{t-1}, w_{t-1}) + \mathrm{cov}(w_t, w_t) + \mathrm{cov}(w_{t+1}, w_{t+1}) \\ &\quad + 2\,\mathrm{cov}(w_{t-1}, w_t)) + 2\,\mathrm{cov}(w_t, w_{t+1})) + 2\,\mathrm{cov}(w_{t-1}, w_{t+1})) \\ &= \tfrac{1}{9}(\sigma_w^2 + \sigma_w^2 + \sigma_w^2 + 0 + 0 + 0) \\ &= \tfrac{3}{9}\sigma_w^2 \\ &= \tfrac{1}{3}\sigma_w^2 \end{aligned} \tag{4}$$

When $s = t + 1$, we have:

$$\begin{aligned} \gamma_v(t+1,t) &= \mathrm{cov}(\tfrac{1}{3}(w_t + w_{t+1} + w_{t+2}), \tfrac{1}{3}(w_{t-1} + w_t + w_{t+1})) \\ &= \tfrac{1}{9}(\mathrm{cov}(w_t, w_t) + \mathrm{cov}(w_{t+1}, w_{t+1})) \\ &= \tfrac{2}{9}\sigma_w^2 \end{aligned} \tag{5}$$

If we then follow this for more lags, we get:

$$\gamma_v(s,t) = \begin{cases} \frac{3}{9}\sigma_w^2, & \text{if } s = t, \\ \frac{2}{9}\sigma_w^2, & \text{if } |s-t| = 1, \\ \frac{1}{9}\sigma_w^2, & \text{if } |s-t| = 2, \\ 0, & \text{if } |s-t| > 2 \end{cases} \tag{6}$$

Why is this interesting? The autocovariance depends only on the *lag* between s and t and not on the absolute location of these time points. We'll come back to this when we talk about *stationarity*.

3.7 Autocovariance of a random walk

Recall that a random walk (with or without drift) is:

$x_t = \delta t + \sum_{i=1}^{t} w_i$

So,

$$\begin{aligned}\gamma(s,t) &= \mathrm{cov}(x_s, x_t) \\ &= \mathrm{cov}(\delta s + \sum_{i=1}^{s} w_i, \delta t + \sum_{i=1}^{t} w_i) \\ &= \sigma^2 \min(s,t)\end{aligned} \tag{7}$$

In this case, the autocovariance *does* depend on the particular s and t chosen.
What if we want a bounded measure?

3.8 Autocorrelation function

We can calculate the autocorrelation function (ACF) as:

$\rho(s,t) = \frac{\gamma(s,t)}{\sqrt{\gamma(s,s)\gamma(t,t)}}$

This measures the linear predictability of the time series at time t (x_t) using x_s. We can also extend this to looking at the linear predictability of one time series to another, by extending into the concepts of *cross-covariance* and *cross-correlation*.

3.9 Cross-covariance

Between two time series x_t and y_t:

$\gamma_x y(s,t) = \mathrm{cov}(x_s, y_t) = \mathbb{E}[(x_s - \mu_{xs})(y_t - \mu_{yt})]$

This tells us how the values in y relate to the values in x over time.

Let's think about a simple example:

$y_t = x_{t-2}$

What is $\gamma_{xy}(k)$ (for lag k)? At what lag is γ_{xy} maximized?

We can also have the normalized version:

3.10 Cross-correlation

$\rho_{xy}(s,t) = \frac{\gamma_x y(s,t)}{\sqrt{\gamma_x(s,s)\gamma_y(t,t)}}$

This is bounded such that $-1 \leq \rho(s,t) \leq 1$

Poll example

4 Next time:

- Stationarity
- Calculating sample correlation/covariance

- Multi-D time series
- RNGs
- Measures of dependence! Read SS Chapter 1, sections 1.3-1.7.